

Introduction to Computer Architecture

Sheet #8

Exercises:

1. What is the difference between using direct and indirect addressing? Give an example.

ANSWER:

Direct addressing provides the actual memory address of the operand in the instruction, whereas indirect addressing provides, as part of the instruction, a pointer to a memory location. For example, the instruction Load X interpreted using direct addressing would go to memory location X and load the value found there. Using indirect addressing, memory location X would be used as the effective address of what should actually be loaded. So if a value of 200 were found at location X, the value located at address 200 would be loaded.

2. Suppose we have the instruction Load 1000. Given that memory and register R1 contain the values below:

Memory	
1000	1400
...	
1100	400
...	
1200	1000
...	
1300	1100
...	
1400	1300

R1
200

Assuming R1 is implied in the indexed addressing mode, determine the actual value loaded into the accumulator and fill in the table below:

Mode	Value Loaded into the AC
Immediate	1000
Direct	1400
Indirect	1300
Indexed	1000

3. Suppose we have the instruction Load 500. Given that memory and register R1 contain the values below:

100	600
...	
400	300
...	
500	100
...	
600	500
...	
700	800

Assuming **R1 contains 100** and is implied in the indexed addressing mode and based addressing mode, determine the actual value loaded into the accumulator and fill in the table below:

Mode	Value Loaded into the AC
Immediate	500
Direct	100
Indirect	600
Indexed	500
Register	100
Register Indirect	600
Based Addressing	500

4. A nonpipelined system takes 200ns to process a task. The same task can be processed in a 5-segment pipeline with a clock cycle of 40ns. Determine the speedup ratio of the pipeline for 200 tasks. What is the maximum speedup that could be achieved with the pipeline unit over the nonpipelined unit?

ANSWER:

$$\text{SpeedUp} = (200\text{ns} \times 200) / ((5+200-1)(40\text{ns})) = 40000/8160 = 4.91$$

$$\text{Max SpeedUp} = 5$$

5. A nonpipelined system takes 300ns to process a task. The same task can be processed in a 5-segment pipeline with a clock cycle of 60ns. Determine the speedup ratio of the pipeline for 100 tasks. What is the maximum speedup that could be achieved with the pipeline unit over the nonpipelined unit?

ANSWER:

$$\text{SpeedUp} = (300\text{ns} \times 100) / ((5+100-1)(60\text{ns})) = 30000/6240 = 4.8$$

$$\text{Max SpeedUp} = 5$$

6. Write code to implement the expression: $A = (B + C) * (D + E)$ on 3-, 2-, 1- and 0-address machines. In accordance with programming language practice, computing the expression should not change the values of its operands.

ANSWER:

Add R1, B, C
Add R2, D, E
Mult A, R1, R2

3-address machine

Load R1, B
Add R1, C
Load R2, E
Add R2, E
Mult R2, R1
Store A, R2

2-address machine

Load B
Add C
Store Temp
Load D
Add E
Mult Temp
Store A

1-address machine

Push B
Push C
Add
Push D
Push E
Add
Mult
Store A

0-address machine

7. The memory unit of a computer has 256K words of 32 bits each. The computer has an instruction format with 4 fields: an opcode field; a mode field to specify 1 of 7 addressing modes; a register address field to specify one of 60 registers; and a memory address field. Assume an instruction is 32 bits long. Answer the following:
- How large must the mode field be?
 - How large must the register field be?
 - How large must the address field be?
 - How large is the opcode field?

ANSWER:

- We need to identify 1 of 7 items, so there must be 3 bits ($2^3 = 8$)
- 60 registers implies 6 bits ($2^6 = 64$)
- $256\text{K} = 2^8 2^{10} = 2^{18}$, or 18 bits
- $32 - (3 + 6 + 18) = 5$ bits